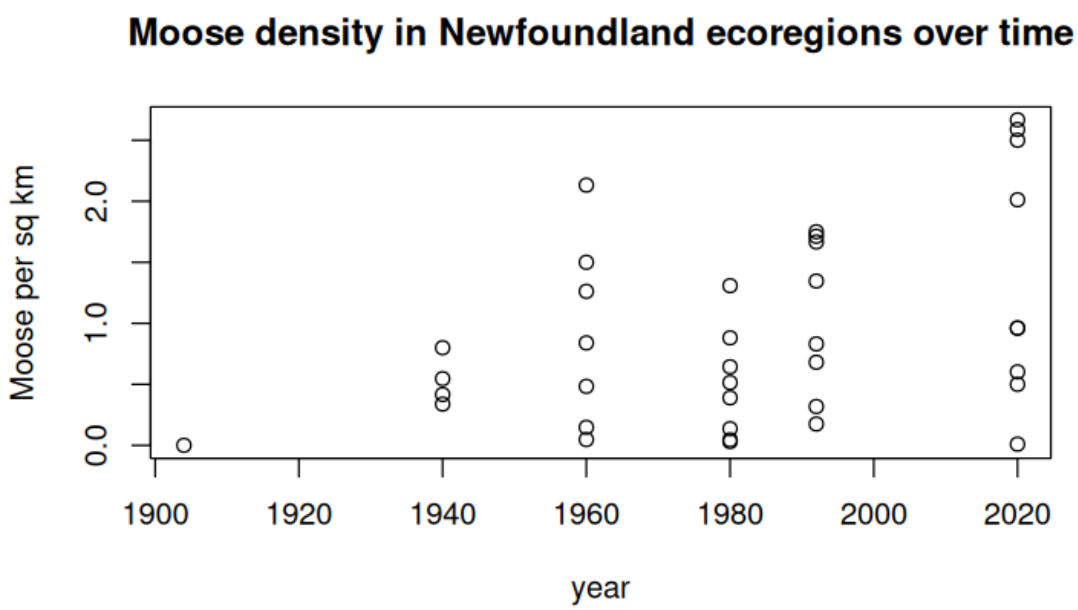
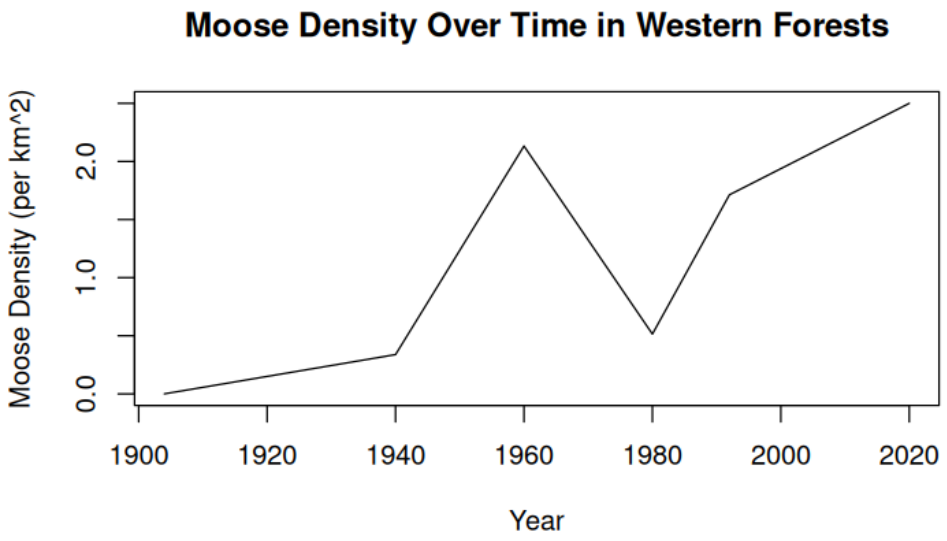
```
install.packages(dplyr)
library(dplyr)
install.packages("dplyr")
library(dplyr)
moosedata <- read.csv("MoosePopulation.csv")
View(moosedata)
moose_clean <- na.omit(moosedata)
moose_sel <- select(moose_clean, Ecoregion, Year, Area,
Estimated_Moose_Pop)
year_min <- min(df$Year)
year_min <- min(moose_sel$Year)
moose_max <- max(Estimated_Moose_Pop, na.rm = TRUE)
moose_max <- max(Estimated_Moose_Pop, na.rm = TRUE)
moose_max <- max(moose_sel$Estimated_Moose_Pop, na.rm = TRUE)
moose_max
plot(moosedata2$Year, moosedata2$MooseDensity,
xlab = "year",
ylab = "Moose per sq km",
main = "Moose density in Newfoundland ecoregions over time")
moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
# Subset the data for the Western_Forests region
west <- subset(moose, Region == "Western_Forests")
# Subset the data for the Western_Forests region
west <- subset(moosedata2, Region == "Western_Forests")
plot(moosedata2$Year, moosedata2$MooseDensity,
xlab = "year",
ylab = "Moose per sq km",
main = "Moose density changed in western regions over time")
moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
main = "Moose density changed in western regions over time)
plot(moose_west$Year,
moose_west$MooseDensity,
type = "l",
xlab = "Year",
ylab = "Moose Density (per km^2)",
main = "Moose Density Over Time in Western Forests")
moose_2020 <- filter(moosedata2, Year == "2020")
moose_2020_high <- moose_2020 %>%
filter(MooseDensity > 2.0)
arrange(moose_2020_high, desc(MooseDensity))
moosefinal <- moosedata2 %>%
filter(Year == 2020) %>%
filter(MooseDensity > 2.0) %>%
arrange(desc(MooseDensity)) %>%
print()
saplings <- read.csv("SaplingStudy.csv")
sap_clean <- na.omit(saplings)
sap_reg_browse <- sap_clean %>%
group_by(Ecoregion) %>%
summarize(mean_browse = mean(BrowsingScore, na.rm = TRUE))
print(sap_reg_browse)
avg_browse_reg <- moosefinal %>%
group_by(Ecoregion) %>%
summarise(AverageBrowsing = mean(AverageBrowsing, na.rm = TRUE)) %>%
```

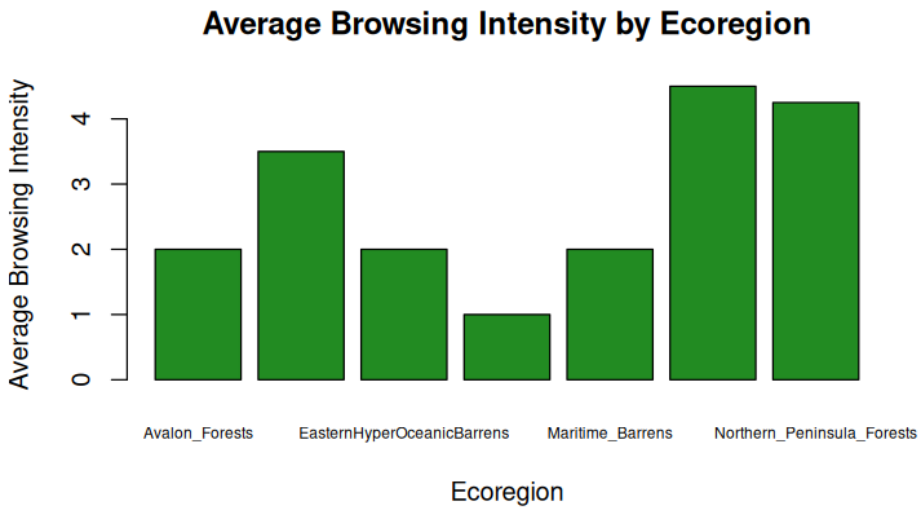
```
arrange(desc(AverageBrowsing))   # Highest at top, lowest at bottom
avg_browse_reg <- moosefinal %>%
group_by(Ecoregion) %>%
summarise(AverageBrowsing = mean(AverageBrowsing, na.rm = TRUE)) %>%
arrange(desc(AverageBrowsing))   # Highest at top, lowest at bottom
rlang::last_trace()
avg_browse_reg <- sap_reg_browse %>%
arrange(desc(AverageBrowsing))
avg_browse_reg <- sap_reg_browse %>%
arrange(desc(BrowsingScore))
avg_browse_reg <- sap_reg_browse %>%
arrange(desc(mean_browse))
print(avg_browse_reg)
View(avg_browse_reg)
# Highest browsing: <Northern_Peninsula_Forests>
# Lowest browsing: <StraitOfBelleIsleBarrens>
sap_reg_height <- saplings %>%
group_by(Ecoregion) %>%
summarize(meanheight = mean(Height, na.rm = TRUE))
print(sap_reg_height)
sap_reg_height_low <- sap_reg_height %>%
filter(meanheight < 20)   # ecoregions with severe browsing (<20 cm)
print(sap_reg_height_low)
View(sap_reg_height_low)
sap_spe_browse <- saplings %>%
group_by(Species) %>%
summarize(mean_browse = mean(BrowsingScore, na.rm = TRUE))
print(sap_spe_browse)
View(sap_spe_browse)
avg_browse_spe <- sap_spe_browse %>%
arrange(desc(mean_browse))
avg_browse_spe
View(avg_browse_spe)
fir_reg_browse <- browse_data %>%
filter(Species == "Balsam_Fir") %>%
group_by(Ecoregion) %>%
summarize(mean_browse = mean(BrowsingScore, na.rm = TRUE))
View(fir_reg_browse)
avg_browse <- tapply(fir_reg_browse$mean_browse,
fir_reg_browse$Ecoregion,
mean)
barplot(fir_reg_browse$mean_browse,
names.arg = fir_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Intensity",
main = "Average Browsing Intensity by Ecoregion",
col = "forestgreen",
cex.names = 0.6)
View(avg_browse_reg)
spruce_reg_browse <- moose_data %>%
spruce_reg_browse <- saplings %>%
filter(Species == "Black_Spruce") %>%
group_by(Ecoregion) %>%
summarise(mean_browse = mean(BrowsingScore, na.rm = TRUE))
```

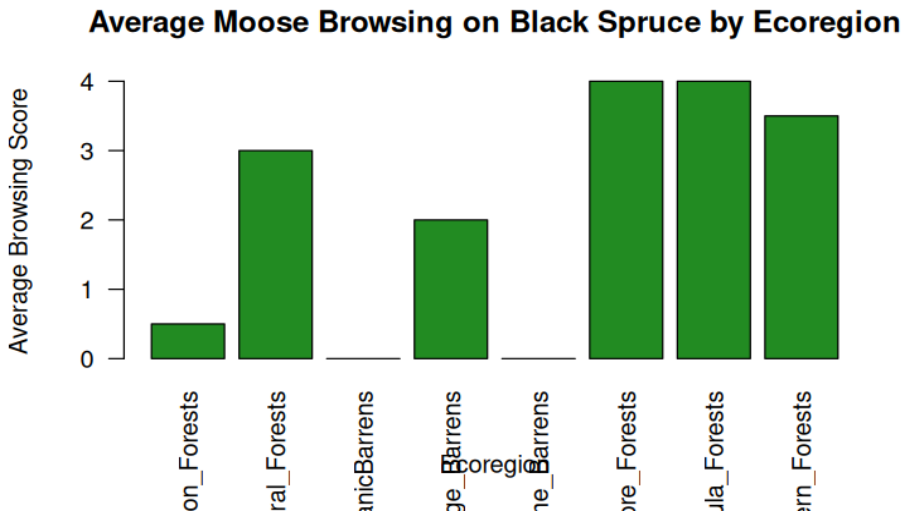
```
barplot(
height = spruce_reg_browse$mean_browse,
names.arg = spruce_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Average Moose Browsing on Black Spruce by Ecoregion",
col = "forestgreen",
las = 2
)
View(spruce_reg_browse)
#Black Spruce browsing tends to be lower and more uniform across
ecoregions, while Balsam Fir browsing is generally higher and shows
stronger regional variation due to its greater palatability.
sap_spe_tally <- sap_clean %>%
group_by(Species) %>%
tally() %>%
print()
# The SaplingStudy dataset is not evenly distributed; some ecoregions and
species appear far more often,
# while others have very few samples, making them underrepresented.
# Recognizing bias in ecological datasets matters because uneven sampling
can distort patterns we think we see
# and lead to inaccurate conclusions about species, habitats, or
ecological processes.
moose_2020b <- moose_clean %>%
filter(Year == 2020) %>%
mutate(MooseDensity = Estimated_Moose_Pop / Area)
View(moose_2020b)
moose_sap <- left_join(moose_2020b, sap_clean, by = 'Ecoregion',
relationship = "many-to-many")
sum_spe_browse <- moose_sap %>%
group_by(Species, Ecoregion) %>%
summarize(
mean_browse = mean(BrowsingScore, na.rm = TRUE),
mean_moose = mean(MooseDensity, na.rm = TRUE)
)

print(sum_spe_browse)
View(sum_spe_browse)
library(ggplot2)
install.packages(ggplot2)
library(ggplot2)
#yes, the evidence generally supports the hypothesis. Moose tend to be
choosy when food is abundant and shift toward generalist browsing when
population density rises.
#moose tend to favour willows and alders the most. They tend to browse
black spruce the least.
#Black ash is not shown on the figure. this is because there is a small
data sample for black ash consisting of 1 entry from the western forests.
collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
human_pop <- c(18000, 12000, 4000, 75100, 24000,3500, 32000, 270000,
2300)
study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests",
"Long_Range_Barrens","Central_Forests","Western_Forests","EasternHyperOce
```

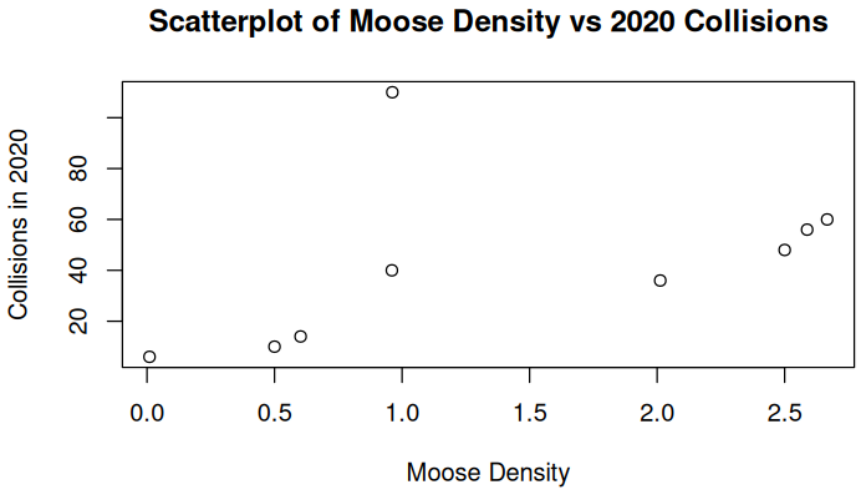
```
anicBarrens","Maritime_Barrens","Avalon_Forests","StraitOfBelleIsleBarren
s")
moose_coll <- data.frame(collisions2020, human_pop, study_sites)
View(moose_coll)
moose_coll2 <- moose_coll %>%
rename_with(~ "Ecoregion", study_sites)
left_join(moose_coll2, moose_2020, by = "Ecoregion")
rename_with(moose_coll = moosecoll2)
rename_with(moose_coll2 = moose_coll)
coll_merge <- left_join(moose_coll2, moose_2020, by = "Ecoregion")
plot(coll_merge$MooseDensity,
coll_merge$collisions2020,
xlab = "Moose Density",
ylab = "Collisions in 2020",
main = "Scatterplot of Moose Density vs 2020 Collisions")
# The scatterplot shows that a higher moose density typically results in
more collisions. However, collisions in the avalon forests region are
much higher even though its moose density is less than 1.
coll_merge_per_capita <- coll_merge %>%
mutate(coll_per_capita = collisions2020 / human_pop)
View(coll_merge_per_capita)
plot(coll_merge_per_capita$human_pop,
coll_merge_per_capita$coll_per_capita,
xlab = "Human Population",
ylab = "Collisions per Capita",
main = "Scatterplot of Collisions per Capita vs Human Population")
# The scatterplot shows that a higher human population results in fewer
collisions. this is becasuse in higher populated area there is less
forest for moose leasing to less moose in the area.
```

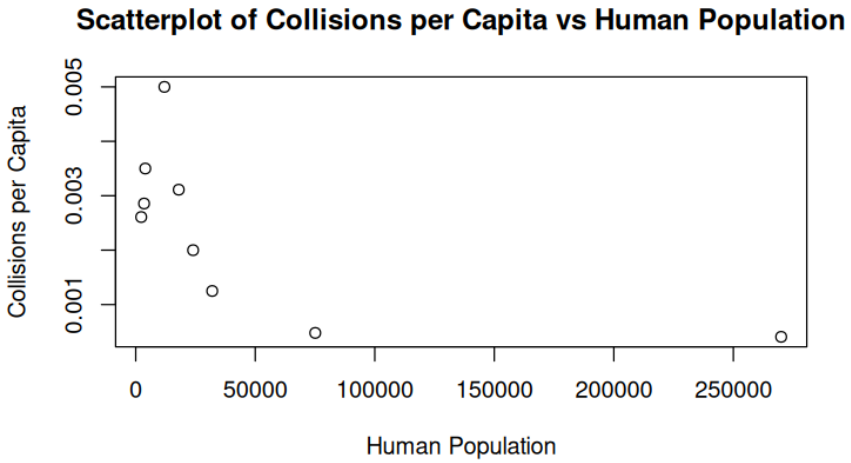



Moose Density Over Time in Western Forests
Moose Density (per km^2)
0.0
1.0
2.0
1900
1920
1940
1960
1980
2000
2020
Year


Average Browsing Intensity by Ecoregion
Average Browsing Intensity
0
1
2
3
4
Avalon_Forests
EasternHyperOceanicBarrens
Maritime_Barrens
Northern_Peninsula_Forests
Ecoregion


Average Moose Browsing on Black Spruce by Ecoregion
Average Browsing Score
0
1
2
3
4
on_Forests
ral_Forests
nicBarrens
ge_Barrens
ne_Barrens
re_Forests
la_Forests
rn_Forests
Ecoregion


Scatterplot of Moose Density vs 2020 Collisions
Collisions in 2020
20
40
60
80
0.0
0.5
1.0
1.5
2.0
2.5
Moose Density


Scatterplot of Collisions per Capita vs Human Population
Collisions per Capita
0.001
0.003
0.005
0
50000
100000
150000
200000
250000
Human Population